

ABHAY SONI

Satna, Madhya Pradesh | +91 6263566876 | soniabhay2008@gmail.com | LinkedIn | GitHub

Summary

3rd-year B.Tech Computer Science student specializing in SOC operations and blue-team security. Hands-on experience in security monitoring, log analysis, alert triage, threat detection, and phishing analysis through two projects: Nexus-SOC, a SIEM-style SOC simulator with MITRE ATT&CK-mapped detection rules and incident response ticketing, and PHISHIQ, an ML-based phishing detector for URLs, emails, and SMS. Tools include Wireshark, Nmap, VirusTotal, AbuseIPDB, and Python.

Education

AKS University, Satna, Madhya Pradesh

B.Tech in Computer Science and Engineering

Aug 2024 – Jul 2028 (Expected)

CGPA: 8.86/10

Projects

Nexus-SOC – Security Operations Center Simulator | GitHub | Live Demo

Aug 2026 – Oct 2026

Python, Flask, SQLite, MITRE ATT&CK

- Built a SIEM-style SOC simulator in Flask and SQLite that ingests security events, parses and normalizes raw logs, and raises alerts using 8 detection rules mapped to MITRE ATT&CK techniques (T1110, T1046, T1078, T1190, T1566).
- Implemented event correlation rules for brute-force attacks (5+ failed logins in 5 minutes) and port scans, with alert deduplication to reduce noise and false positives.
- Developed IOC extraction (IPs, domains, URLs, emails, MD5/SHA1/SHA256) and threat intelligence enrichment that automatically raises alert severity on indicator matches.
- Automated the incident response workflow with ticketing, severity-based priority (P1–P4), status tracking, and PDF/CSV reporting; added a live attack simulator and analyst dashboard.
- Secured the application with session authentication, CSRF protection, rate limiting, and per-user data isolation; wrote 25 pytest unit tests covering detection, IOC, and log parsing logic.

PHISHIQ – Multi-Channel Phishing Detection | GitHub

Mar 2026 – Jun 2026

Python, Flask, Scikit-learn, Tesseract OCR

- Built a Flask web app for phishing analysis that scores URLs, emails, and SMS messages by blending machine learning models with a rule engine (final score = $0.4 \times \text{rule} + 0.6 \times \text{ML}$) and explains each verdict with the top 5 ranked reasons.
- Engineered 36 lexical URL features and trained a calibrated Random Forest (300 trees) using a domain-grouped train/test split to prevent leakage; achieved 98.63% accuracy and 0.9967 ROC-AUC on 42,069 held-out URLs.
- Trained TF-IDF + Logistic Regression models for email (99.8% accuracy) and a 3-class SMS model (ham/spam/smishing, 97.69% accuracy), with India-specific scam rules for KYC, UPI, and OTP-sharing messages.
- Added threat detection rules for @ in URLs, IP hosts, punycode, typosquatting, and brand impersonation, plus trusted-domain handling to reduce false positives; integrated Tesseract OCR to scan screenshots.
- Wrote 12 pytest test cases covering feature extraction, risk thresholds, score combination, and email/SMS rule engines.

Technical Skills

SOC Operations: Alert Triage, Log Analysis, Event Correlation, Detection Rules (Brute-Force, Port-Scan), IOC Analysis, Threat Intelligence Enrichment, Incident Ticketing, Phishing Analysis, False Positive Reduction

Frameworks: MITRE ATT&CK, Incident Response Lifecycle

Security Tools: Wireshark, Nmap, VirusTotal, AbuseIPDB, Windows Event Viewer, Linux CLI

Networking: TCP/IP, DNS, HTTP/S, Packet Analysis, Network Scanning

Programming: Python (log parsing, API lookups), Flask, SQLite, PowerShell (basics)

Application Security: OWASP Top 10 (Fundamentals), Session Authentication

Certifications

- Introduction to Operating Systems (Elite) – NPTEL, IIT Madras

Jul – Sep 2026